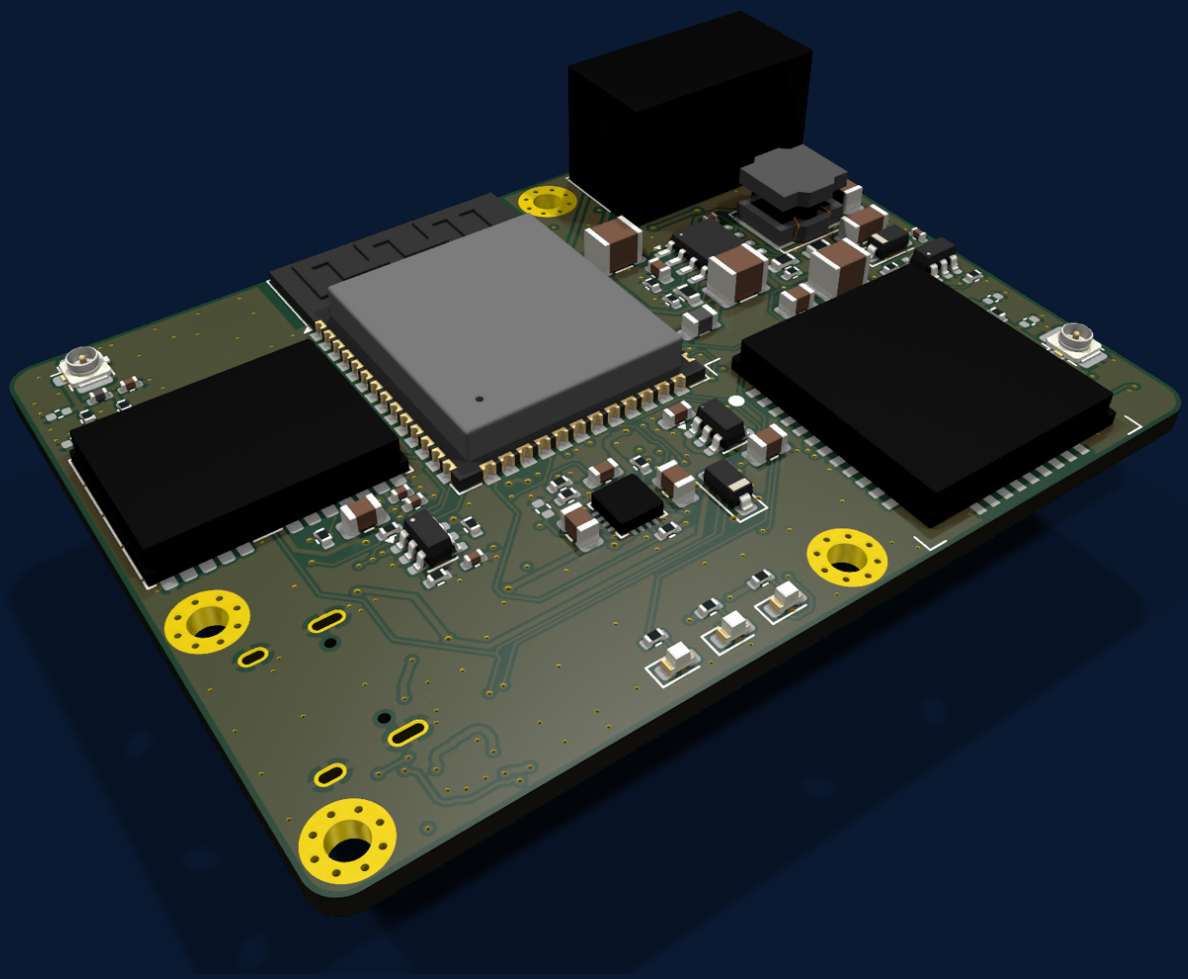
SAMAY ELECTRONICS

Samay Track 2W

Product, design, cost and business plan



₹6,183 hardware cost / unit	**₹26–50 lakh** to reach the first 100 sales	**≈ 10–12 months** to a production launch	**Rev A0** PCB designed, not built yet

All prices are planning estimates (October 2026, INR, excluding GST). Confirm them with supplier quotes, test labs and a compliance consultant before you commit money.

Contents

#	Section
1	Executive summary
2	The product and how it works
3	Hardware design
4	BOM cost: line by line
5	Unit economics and profit margins
6	Lite version: design and cost
7	Competitors and our difference
8	App and cloud: design and cost
9	Certifications and legal
10	AIS-140: what, who, cost
11	How to sell (and Amazon)
12	Total money needed (investment plan)
13	Timeline: idea to market
14	What you were missing, and risks
15	Next 10 steps

1. Executive summary

Topic	Answer in one line
Product	Premium 4G LTE-M GPS + NavIC tracker for motorcycles and scooters, with a 40 h backup battery, crash/theft/SOS alerts and an engine lock
Status	PCB fully designed (DRC/ERC clean, Gerbers, BOM). **No prototype yet, no firmware, no app.**
Unit cost	≈ ₹6,183 hardware at 1,000 pcs (≈ ₹6,600 landed). Lite version ≈ ₹1,200–1,500
Price	Rider ₹10,999 · Fleet ₹7,999 · Renewal ₹1,999/yr (suggested; the draft poster still shows ₹9,499 / ₹7,499). Lite ₹2,999–3,499
Margin	≈ 31 % direct, 13 % via dealers, 18 % fleet. Renewals ≈ 70 %: **the profit is in subscriptions**
Best market	B2B first: two-wheeler loan companies, rentals, delivery fleets, small EV makers. Then dealers and Amazon (Lite)
Money needed	≈ ₹26–50 lakh to the first 100 paid units. Add inventory for volume, and ₹15–25 lakh + guarantees if you do AIS-140
Time	≈ 6–8 months to the pilot, 10–12 months to a production launch
Biggest gaps	Firmware, a working prototype, enclosure, app/server, certificates, SIM contract, funding

2. The product and how it works

The tracker sits under the seat. It reads satellites (GPS, Galileo, BeiDou and India's NavIC) and senses motion. It sends position and alerts over the mobile network to your cloud server, which shows them in the app. **There is no distance limit**: it works anywhere with LTE-M / NB-IoT coverage, and stores data offline where there is no coverage.

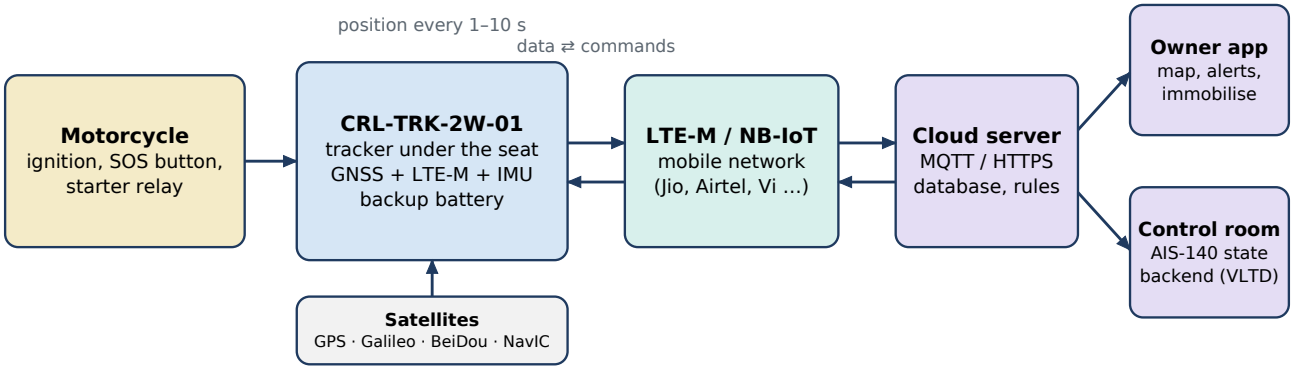


Figure 1: the system, from the bike to the people who need the information.

Key specifications

Item	Value
Positioning	u-blox NEO-F10N, dual-band L1 + L5, GPS / Galileo / BeiDou / QZSS / **NavIC**; ≈ 2 m in open sky
Network	SIMCom SIM7080G LTE Cat-M1 / NB-IoT, nano-SIM (eSIM in rev B), no 2G
Brain	ESP32-S3: BLE setup, TLS, OTA updates, rules
Sensors	IIM-42652 6-axis IMU (wake-on-motion, crash), RTC with tamper input, battery gauge
Memory	16 MB flash ≈ 4 lakh positions offline
Power	9–36 V vehicle input, reverse and load-dump protected; 18650 backup ≈ 40 h tracking; sleep ≈ 0.11 mA
I/O	Ignition, SOS, DIN1, AIN1 (0–30 V), DOUT1 relay (engine lock), USB-C service
Size	PCB 60 × 45 mm, 4-layer; IP67 enclosure target
Report rate	every 10 s while driving (configurable); heartbeat while parked

3. Hardware design

Designed in KiCad 9 from Python source files, so every output can be rebuilt. The schematic has 6 sheets and the PCB has 180 parts on 4 layers, with parts on both sides. Checks: **DRC 0 errors, ERC clean, schematic and PCB in sync**. The outputs are ready for JLCPCB / PCBWay: Gerbers, BOM, pick-and-place, 3D STEP, assembly drawings.

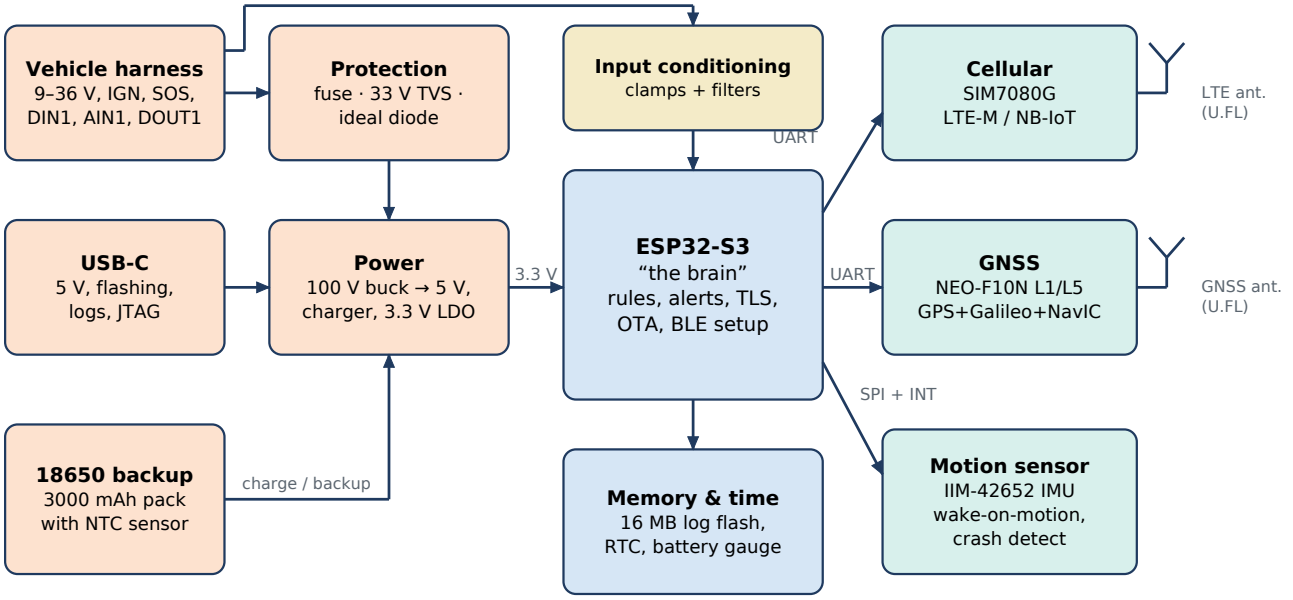
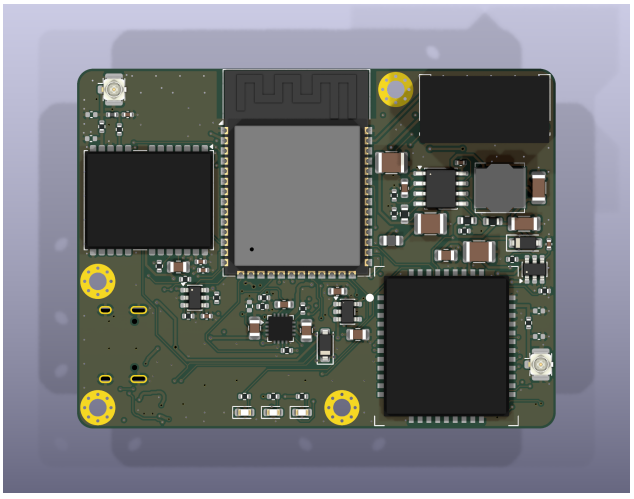
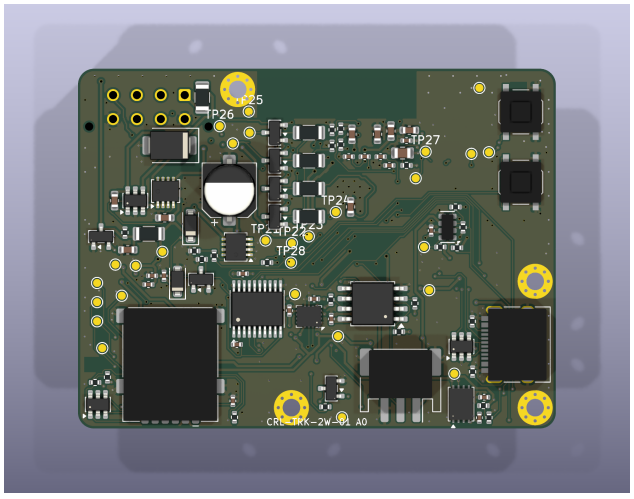


Figure 2: functional block diagram.



Top side: modules, power, connectors



Bottom side: small parts, SIM, USB-C, test pads

Design files in the repository

Need	File
Schematic (PDF)	04_docs/CRL-TRK-2W-01_schematic.pdf
KiCad project (Altium imports it)	01_kicad/tracker.kicad_pro
Manufacturing	03_fab/ Gerbers zip, BOM_JLCPCB.csv, CPL_JLCPCB.csv
Engineering report	04_docs/DESIGN_REPORT.md (power, EMC, SI, DFM, test)
Values to check before ordering	04_docs/VERIFY_LIST.md

4. BOM cost, line by line

Estimated at 1,000 pieces with typical distributor prices (₹88 per USD). Every line is in **03_fab/CRL-TRK-2W-01_BOM_costed.csv**. Three modules make up two-thirds of the board cost.

Total ≈ ₹6,183 per unit at 1,000 pcs

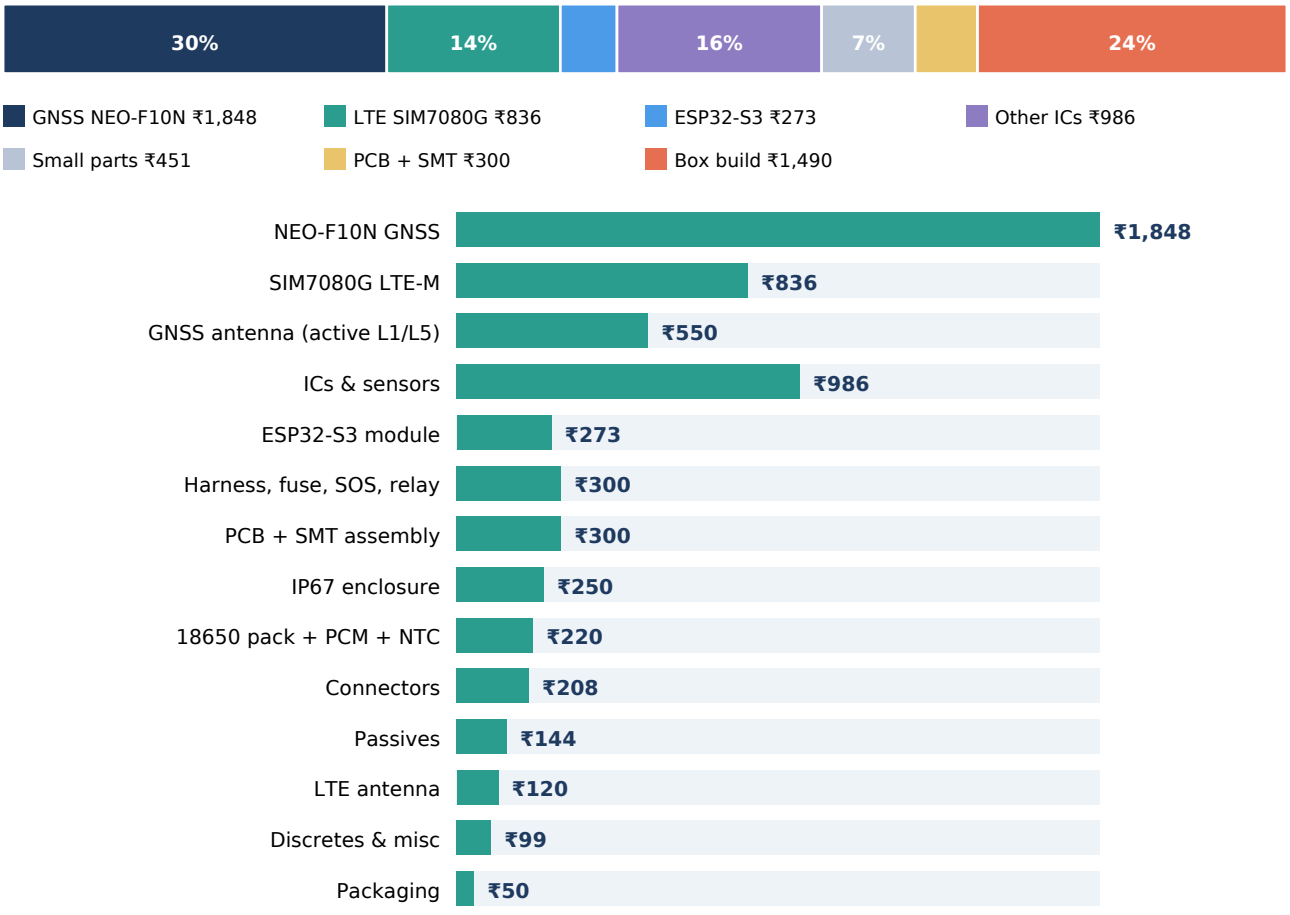


Figure 3: cost per unit by item (₹, 1,000 pcs).

Quantity	Hardware / unit	Why
10 (prototype)	≈ ₹9,000–12,000	small-quantity prices, setup fees, courier
100 (pilot)	≈ ₹7,500	partial price breaks
1,000	≈ ₹6,183 (≈ ₹6,600 landed)	the basis of this plan
10,000	≈ ₹5,300–5,600	module prices fall 15–20 % at this volume

Cost-down options: ICM-42670 IMU instead of IIM-42652 (≈ ₹200 saved) · generic U.FL connectors · combine buck + LDO · negotiate modules at volume · MAX-F10S or a cheaper NavIC GNSS once it is qualified.

5. Unit economics and profit margins

Costs per unit, beyond the hardware: year-1 data and cloud ₹600 and installation ₹300 (Rider only; fleets pay data through the platform fee), warranty reserve 3 % of price, packaging and shipping ₹150. Dealer margin is 18 % of price. Fixed costs such as staff and certification are covered in section 12.

Plan	Price	Total cost	Profit / unit	Margin
Rider: direct (poster)	₹9,499	₹7,518	₹1,981	21 %
Rider: dealer (poster)	₹9,499	₹9,228	₹271	3 %
Fleet (poster)	₹7,499	₹6,558	₹941	13 %
Rider: direct (suggested)	₹10,999	₹7,563	₹3,436	31 %
Rider: dealer (suggested)	₹10,999	₹9,543	₹1,456	13 %
Fleet (suggested)	₹7,999	₹6,573	₹1,426	18 %
Renewal / year	₹1,999	₹600	₹1,399	70 %
Fleet platform fee / year	₹1,499	₹600	₹899	60 %

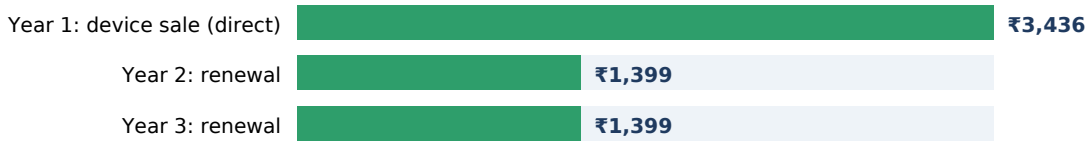


Figure 4: profit from one direct Rider customer over 3 years ≈ ₹6,234.

Conclusion: at the poster prices (₹9,499 / ₹7,499) hardware margins are too thin, especially through dealers (≈ 3 %). Use the **suggested prices** (₹10,999 / ₹7,999) and push yearly renewals. Every renewing customer adds ≈ ₹1,400 a year at about 70 % margin.

6. Lite version: design and cost

A cheaper model for consumers and Amazon. It is a **new 2-layer board**: one 4G Cat-1 module with 2G fallback runs the firmware itself, so the ESP32 is dropped, and the GNSS is a cheaper single-band part.

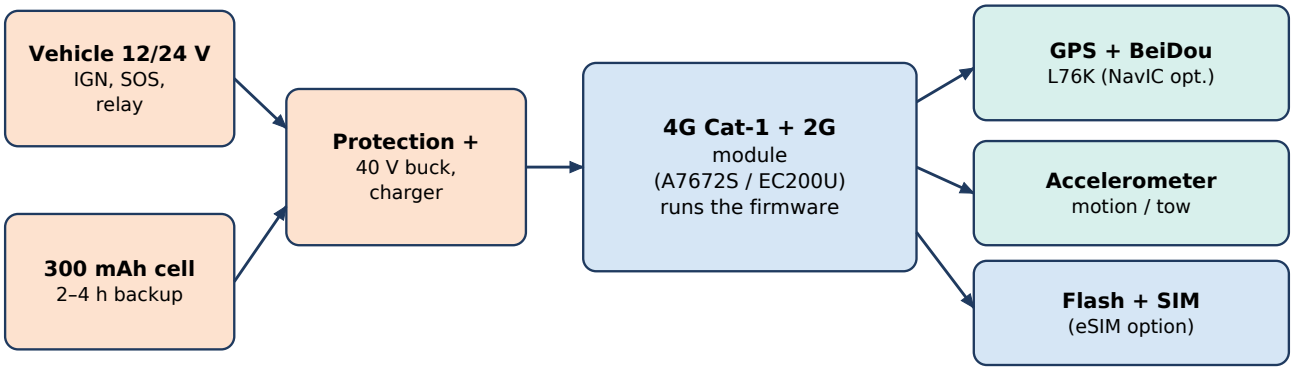


Figure 5: Lite block diagram.

Part	Choice	₹ / unit
4G modem	Cat-1 + 2G (SIMCom A7672S / Quectel EC200U), also runs the app firmware	500–650
GNSS	L76K-class GPS + BeiDou (NavIC-capable part +₹300–400)	200–250
Sensors and power	accelerometer, 40 V buck, charger, 300 mAh cell	140–190
Antennas	on-board GNSS patch + FPC LTE	60–80
Other	flash, SIM, ESD, passives	80–100
PCB + SMT	2-layer, single-sided	80–100
Box build	IP65 enclosure, 4-wire harness, packaging	130–170
Total		**≈ 1,200–1,500**

	Full (Samay Track 2W)	Lite
Hardware cost	≈ ₹6,183	≈ ₹1,200–1,500
Retail price	₹10,999 (suggested)	₹2,999–3,499 incl. year 1
Margin (direct)	≈ 31 %	≈ 35–40 %
NavIC / dual band	yes / yes	optional / no
Backup battery	≈ 40 h	2–4 h
Crash detection, big black box	yes	basic / no
2G fallback	no	yes
AIS-140 possible	yes (rev B with eSIM)	no
Where to sell	fleets, NBFCs, premium riders	Amazon, Flipkart, dealers

7. Competitors and our difference

Segment	Examples	Price (web listings, Oct 2026)
Budget 4G bike trackers	SeTrack, Ajjas 4G Lite, Onelap Micro / LocoShield, Fleettrack 4G	₹900–2,900 incl. year 1, then ≈ ₹999/yr
Mid-range	Alike GPS VT05R	from ₹3,999
AIS-140 VLTD	Atlanta WP 30L, Gtropy, Algotrack and others	₹4,500–13,000, mostly ₹5–8k + install + ₹1–2.5k/yr
Premium imported	Monimoto 9	≈ ₹47,000
Built-in on EVs	Ather, Ola, TVS iQube, Bajaj Chetak	bundled with the scooter

Where Samay Track 2W is different

Samay Track 2W	Typical competitor
Dual-band L1 + L5 GNSS **+ NavIC**	single-band GPS, often no NavIC
≈ 40 h backup battery (18650)	minutes to a few hours, or none
Crash and fall detection (industrial IMU)	vibration alert only
16 MB black box ≈ 4 lakh points	small buffer, gaps in the route
Reverse wiring / 36 V spike proof, 9–36 V	basic fuse
Sleep ≈ 0.11 mA, protects the bike battery	often 2–10 mA
AIS-140 features in a 60 × 45 mm box	large boxes built for buses and trucks

Not different: 4G, live map, app, engine lock, SOS, geofence. Everyone has these. **Weaker:** price, no 2G, not certified yet, LTE-M rural coverage. Verify competitors' spec sheets before you print comparison claims.

Sources: amazon.in listings; kingswayaccessories.com; alikegps.com; sahajgps.com; blog.fleetx.ai; indiamart.com and tradeindia.com AIS-140 listings.

8. App and cloud: design and cost

One Flutter code base builds both the Android and the iPhone app. The tracker never talks to the phone directly; everything goes through your server. A faster route is to start from the open-source **Traccar** server and add your own app on top.

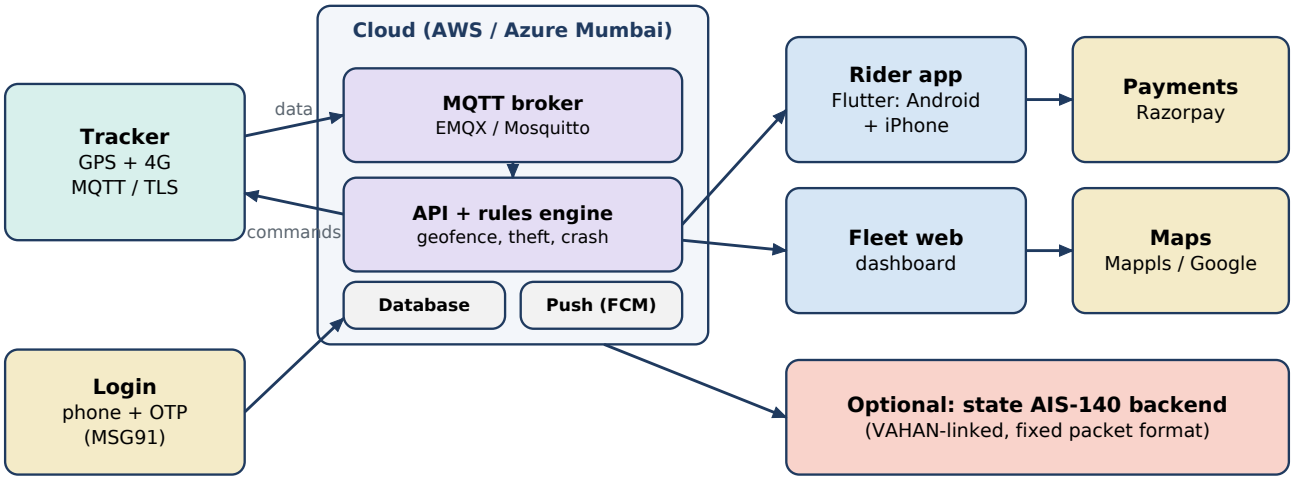


Figure 6: app and cloud architecture.

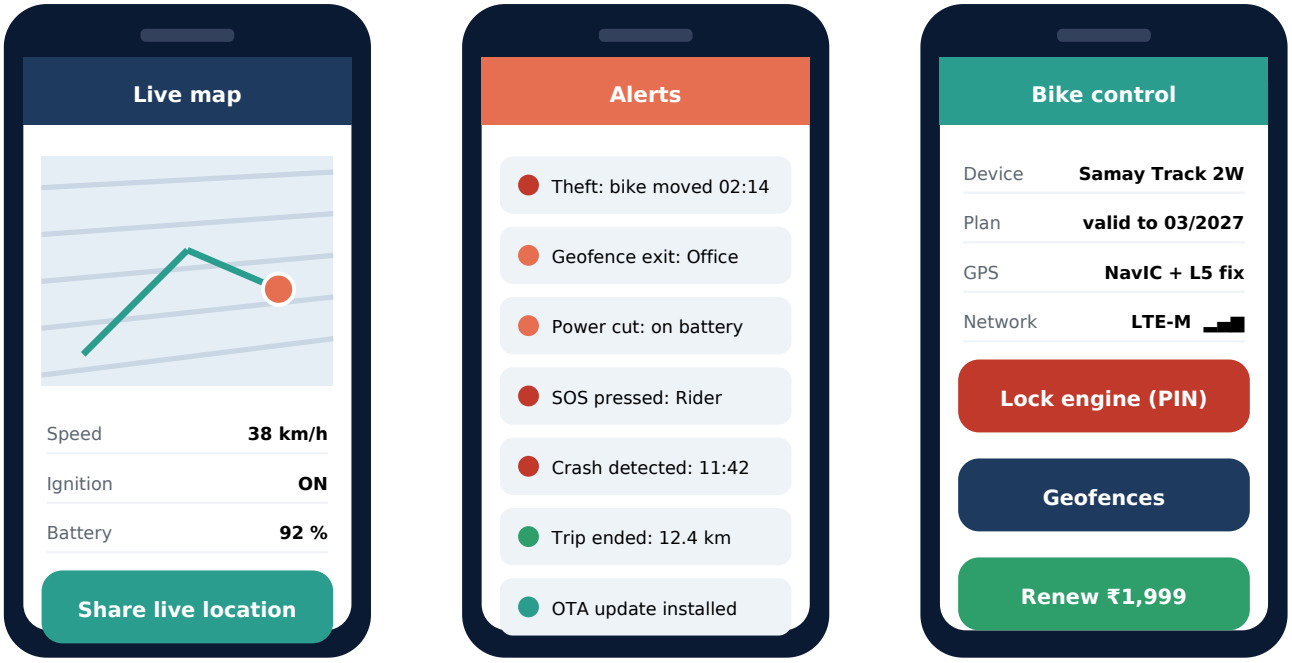


Figure 7: app concept: live map, alerts, bike control (wireframe, not a final UI).

Item	Cost	Notes
Version 1: Traccar + custom Flutter app	₹2–4 lakh, 1–2 months	fastest route to a working demo
Fully custom app + server	₹5–10 lakh, 3–5 months	own protocol, own dashboard
Cloud server	₹3,000–10,000 / month	first few thousand devices
Store accounts	Google $25 once · Apple $99 / yr	iPhone build needs a Mac + Xcode
Maps, OTP SMS, push	≈ ₹1–5 / user / yr	Mappls / Google, MSG91, Firebase (free)
Payments	Razorpay 2 %	for renewals

9. Certifications and legal (before you sell anything)

These apply to **every** version, even without AIS-140. Confirm the exact list with a compliance consultant; the costs are rough estimates.

What	Why	Rough cost	Time
WPC ETA	radio device (LTE, BLE): mandatory	₹0.5–1.5 lakh	1–2 months
BIS (IS 16046) for the Li-ion cell / pack	compulsory registration	₹1–2 lakh (or buy a certified pack)	2–3 months
TEC / MTCTE	may apply to 4G tracking devices: check	₹1–4 lakh	2–4 months
CPCB EPR: e-waste + battery	producer responsibility	₹0.3–1 lakh + yearly targets	1–2 months
DoT M2M service provider registration	if you sell the SIM / data with the device	≈ ₹0.1–0.5 lakh	1–2 months
Pre-compliance EMC, temperature, vibration, ISO 7637	avoid failing the formal tests	₹1–3 lakh	2–4 weeks
Legal Metrology labelling	MRP, maker, support on the box	small	—
DPDP Act: privacy policy, consent, data in India	you store location data	₹0.3–1 lakh (lawyer)	2–4 weeks
Trademark "Samay" + company + GST	brand protection, Amazon Brand Registry	₹0.2–0.5 lakh	filing: days; grant: months
Product liability insurance	engine lock / crash features carry risk	₹0.3–1 lakh / yr	—

Total for basic certificates and legal ≈ ₹5–12 lakh, taking 3–6 months, with several running in parallel. AIS-140 is extra (next section).

10. AIS-140: what, who, cost

AIS-140 is India's standard for **vehicle location tracking devices (VLTD) in public and commercial vehicles**. It is issued under MoRTH by ARAI.

Topic	Details
Must have	GPS + NavIC · panic button linked to the state control room / 112 · live data to the state backend · tamper and power-cut alerts · backup battery · store-and-forward + SMS · embedded multi-operator SIM · OTA
Who must fit it	buses, taxis, cabs, school vans, trucks with permits
Two-wheelers	**exempt** in the 2018 MoRTH notification, as are 3-wheelers and e-rickshaws. Bike-taxi rules vary by state, so check
Approval	type approval by ARAI / ICAT, then a listing in each state

Cost to get AIS-140

Item	Cost	Basis
Type-approval testing (ARAI / ICAT), 3–6 months	≈ ₹5–15 lakh	estimate
Yearly Conformity of Production audit	≈ ₹2–5 lakh / yr	estimate
State registration	₹10,000 + ₹10,000 per model	Sikkim / West Bengal documents
Performance bank guarantee	up to ₹25 lakh per state (refundable)	Sikkim SOP
Integration with each state backend	≈ ₹1–3 lakh per state	estimate
Board rev B: eSIM, 2G question	≈ ₹2–5 lakh	estimate
Total	**≈ ₹15–25 lakh + ₹10–25 lakh+ locked in guarantees**	for 1–2 states

Advice: skip AIS-140 for the launch. Do it only once a fleet or commercial customer is ready to buy. Two-wheeler consumer sales do not need it.

11. How to sell (and Amazon)



Buyer	Why they buy	Offer
Two-wheeler loan companies (NBFCs, banks)	track and recover bikes on unpaid loans	per-unit price + yearly plan, bulk
Bike rental firms	theft protection, renter tracking	free 10-unit pilot, then bulk
Delivery / bike-taxi fleets	rider safety, SOS, route proof	fleet dashboard
Small EV makers	need tracking, no in-house team	white-label ₹1,500–3,000 (Lite) in bulk
Insurers	fewer theft claims	premium discount partnership
Showrooms / accessory shops	extra margin, installs	15–20 % margin, demo unit, poster
Website + WhatsApp	best margin	cash on delivery + installation in your city
Amazon / Flipkart	reach, trust	Lite with a self-install video

Amazon checklist

Need	Detail
Seller account	GSTIN, PAN, bank account; FBA or Easy Ship
Brand Registry	trademark filed for "Samay"
Documents Amazon may ask for	WPC ETA, BIS for the battery, invoices, EPR
Fees	≈ 12–20 % of price (commission, closing, shipping) + 5–10 % ads at launch
Result at ₹2,999 (Lite)	≈ ₹500–1,000 profit per unit before ads

12. Total money needed (investment plan)

Not covered in the earlier notes: how much cash the whole project needs before it earns. Phase 1 takes the product to the first 100 paying units. Phase 2 is volume production.

Phase 1 item	Low (₹ lakh)	High (₹ lakh)
Prototypes: 2 rounds × 10 boards	1.5	2.5
Firmware + app + server (dev, testing)	3	8
Enclosure design + injection mould	3	6
Production test fixture (pogo bed)	0.5	1.5
Pre-compliance testing	1	3
Certificates + legal (section 9)	3	8
Company, trademark, insurance	0.5	1
Pilot batch: 100 units × ≈ ₹7,500	7.5	7.5
Launch marketing (demos, video, ads)	2	5
6 months running costs: cloud, SIM, support	4	8
Phase 1 total	**26**	**50.5**

Phase 2 (choose)	Cash needed	Comment
500 full units inventory	≈ ₹31–33 lakh	modules need 8–12 weeks lead time and are paid upfront
1,000 full units inventory	≈ ₹62–66 lakh	price-break volume
1,000 Lite units	≈ ₹12–15 lakh	after the Lite design (≈ ₹3–5 lakh extra development)
AIS-140 (optional)	≈ ₹15–25 lakh + guarantees	only with a committed fleet customer

Break-even: with a mix of 40 % direct and 60 % dealer sales at ₹10,999, the device profit averages ≈ ₹2,248 per unit. Recovering ₹40 lakh of fixed costs needs **≈ 1,780 units**, or fewer once renewals (≈ ₹1,400 per customer per year) start to add up.

13. Timeline: idea to market

Firmware and the prototype run in parallel. The certificates start once the enclosure and the hardware are frozen. The **critical path** is firmware → validation → certificates → pilot.

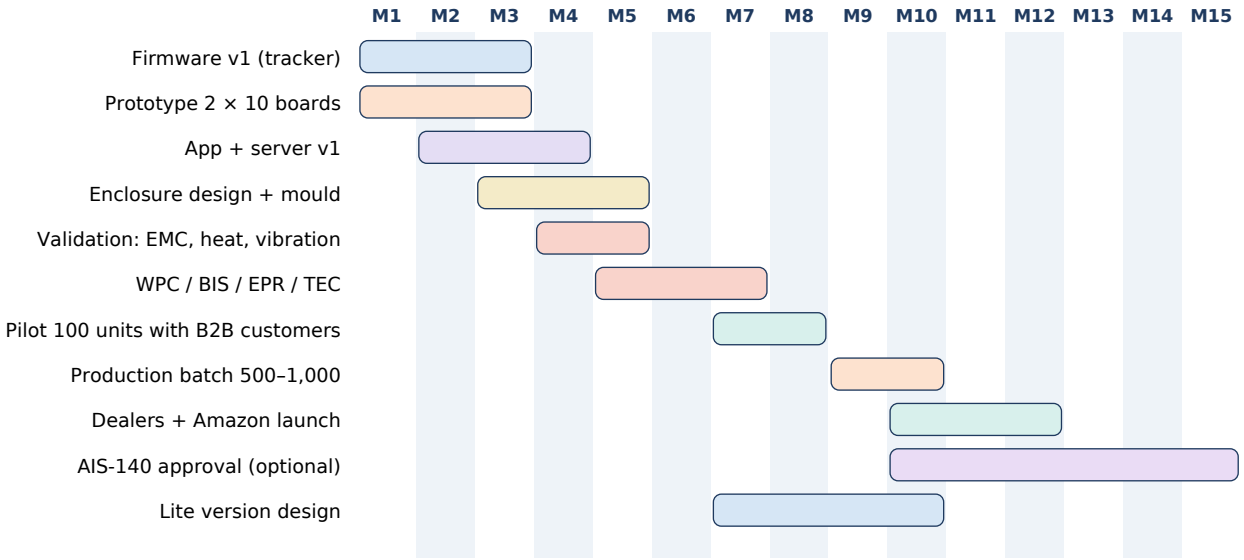


Figure 8: 15-month plan (M1 = month 1).

Milestone	Month	Done when
First boards power up	M2	all rails OK, modules answer (production checklist §8)
End-to-end demo	M4	position on the phone, theft alert, engine lock
Design freeze	M5	validation passed, enclosure and BOM fixed
Certificates in hand	M7	WPC, BIS, EPR (TEC if required)
Pilot customers live	M8	100 units in the field, feedback collected
Production launch	M10–M12	batch built, dealers and Amazon listings live

14. What you were missing, and risks

These items are needed for a real product. They had not been discussed yet, or are not done yet.

#	Missing item	Why it matters	Action
1	**Tracker firmware**	the board does nothing without it	write firmware v1 (Claude can do this)
2	**Built and tested prototype**	the design is not proven; some values are unverified (VERIFY_LIST)	order 10 boards, run the test checklist
3	**Enclosure design + mould**	IP67, mounting, antenna placement	3D design, then a mould ₹3–6 lakh
4	**App + server**	customers see only the app	Traccar + Flutter v1
5	**IoT SIM contract**	data cost per device, eSIM, roaming between operators	talk to Jio / Airtel IoT teams
6	**Warranty, returns, service**	trust, dealer support	1-year replace policy, RMA process
7	**Installer network**	wiring needs skill	train 2–3 installers per city, install video
8	**Customer support**	alerts at night, app issues	WhatsApp support, FAQ, help line
9	**Funding / cash flow**	inventory is paid 2–3 months before sales	plan ₹26–50 lakh + inventory
10	**Team**	firmware, app, sales, installs	1 developer + 1 sales + installers (freelance OK)
11	**Second source for key parts**	NEO-F10N / SIM7080G shortages	qualify alternatives, buffer stock
12	**Engine-lock safety and legal terms**	liability if misused	safe-stop logic, T&C, insurance
13	**Police / data requests policy**	location data is sensitive	DPDP-compliant data policy
14	**Trademark search for "Samay"**	name may already be taken in this class	search before you print
15	**Price validation**	₹10,999 is untested	ask pilot customers, A/B offers

Top risks

Risk	Impact	Mitigation
Price competition from ₹1,500 trackers	high	sell value to B2B; Lite version for consumers
LTE-M / NB-IoT coverage gaps, no 2G	medium	offline log; Lite has Cat-1 + 2G; test routes first
Certification delays	medium	pre-compliance tests, consultant, start early
Component lead time / price	medium	buffer stock, second sources
First-batch field failures	high	pilot with 100 units before volume, OTA fixes
Cash running out before sales	high	B2B pilots with advance payment, phase the spending

15. Next 10 steps

#	Step	Who	Cost
1	Check VERIFY_LIST values, order 10 prototype boards (JLCPCB)	you + Claude	≈ ₹1–1.2 lakh
2	Write tracker firmware v1 + a GPS simulator	Claude	—
3	Build the server (Traccar) + Flutter Android app v1	Claude + you (accounts)	accounts ≈ ₹10k
4	Bring up and test the boards with the production checklist	you / technician	lab time
5	Enclosure: 3D-print a prototype, then quote a mould	mechanical designer	₹0.5 lakh → ₹3–6 lakh
6	Register company / GST, file the trademark, get an IoT SIM deal	you	≈ ₹0.5 lakh
7	Pre-compliance tests, then WPC / BIS / EPR applications	consultant	≈ ₹5–12 lakh
8	Sign 2–3 B2B pilot customers (NBFC, rental, fleet)	you (sales)	demo units
9	Pilot 100 units, collect feedback, fix through OTA	team	≈ ₹7.5 lakh
10	Decide: production batch size, Lite version, AIS-140	you	see section 12

Launch poster (draft)



Related files: 07_marketing/BUSINESS_PLAN_NOTES.md, PRICING_NOTES.md, poster; 04_docs/DESIGN_REPORT.md, CRL-TRK-2W-01_Workflow_UseCases.pdf; 03_fab/CRL-TRK-2W-01_BOM_costed.csv.